

Theorem 2.4. *The following are logical equivalences.*

- (a) $(\phi \rightarrow \psi) \equiv (\neg\phi \vee \psi) \equiv \neg(\phi \wedge \neg\psi) \equiv (\neg\psi \rightarrow \neg\phi)$
- (b) $(\phi \iff \psi) \equiv ((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))$

Proof of (a). We prove the three adjacent equivalences in turn and then chain them.

Claim 1: $(\phi \rightarrow \psi) \equiv (\neg\phi \vee \psi)$.

Let v be any truth assignment.

($\Rightarrow$) Suppose $v((\phi \rightarrow \psi)) = F$. Then $v(\phi) = T$ and $v(\psi) = F$, so that $v(\neg\phi) = F$ and hence $v((\neg\phi \vee \psi)) = F$.

($\Leftarrow$) Suppose $v((\neg\phi \vee \psi)) = F$. Then $v(\neg\phi) = F$ and $v(\psi) = F$, so that $v(\phi) = T$ and hence $v((\phi \rightarrow \psi)) = F$.

Since v takes only the values T and F , the contrapositive of the forwards direction proves

$$v((\neg\phi \vee \psi)) = T \Rightarrow v((\phi \rightarrow \psi)) = T,$$

while the contrapositive of the backwards direction proves

$$v((\phi \rightarrow \psi)) = T \Rightarrow v((\neg\phi \vee \psi)) = T.$$

Therefore $v((\phi \rightarrow \psi)) = F \iff v((\neg\phi \vee \psi)) = F$ and $v((\phi \rightarrow \psi)) = T \iff v((\neg\phi \vee \psi)) = T$, so that $v((\phi \rightarrow \psi)) = v((\neg\phi \vee \psi))$ for every truth assignment v . Thus $(\phi \rightarrow \psi) \equiv (\neg\phi \vee \psi)$.

Claim 2: $(\neg\phi \vee \psi) \equiv \neg(\phi \wedge \neg\psi)$.

Theorem 2.3(h) provides us with the following De Morgan's Law: $\neg(\phi \wedge \psi) \equiv (\neg\phi \vee \neg\psi)$. Treating the formulas in Theorem 2.3(h) as metavariables and substituting $\phi \mapsto \phi$, $\psi \mapsto \neg\psi$ uniformly on both sides, we arrive at

$$\neg(\phi \wedge \neg\psi) \equiv (\neg\phi \vee \neg\neg\psi).$$

Now Theorem 2.3(g) states the law of double negation: $\neg\neg\phi \equiv \phi$. Substituting $\phi \mapsto \psi$ gives $\neg\neg\psi \equiv \psi$, so for every truth assignment v we have $v(\neg\neg\psi) = v(\psi)$. By the truth table for $\vee$ it follows that $v((\neg\phi \vee \neg\neg\psi)) = v((\neg\phi \vee \psi))$ for every v , so that

$$(\neg\phi \vee \neg\neg\psi) \equiv (\neg\phi \vee \psi).$$

Chaining the two displayed equivalences by Exercise 2.35(c) and reversing the result by Exercise 2.35(b), we conclude that $(\neg\phi \vee \psi) \equiv \neg(\phi \wedge \neg\psi)$.

Claim 3: $\neg(\phi \wedge \neg\psi) \equiv (\neg\psi \rightarrow \neg\phi)$.

Let v be any truth assignment.

($\Rightarrow$) Suppose $v(\neg(\phi \wedge \neg\psi)) = F$. Then $v((\phi \wedge \neg\psi)) = T$, so $v(\phi) = T$ and $v(\neg\psi) = T$, so that $v(\neg\phi) = F$ and hence $v((\neg\psi \rightarrow \neg\phi)) = F$.

($\Leftarrow$) Suppose $v((\neg\psi \rightarrow \neg\phi)) = F$. Then $v(\neg\psi) = T$ and $v(\neg\phi) = F$, so that $v(\phi) = T$, hence $v((\phi \wedge \neg\psi)) = T$ and $v(\neg(\phi \wedge \neg\psi)) = F$.

Since v takes only the values T and F , the contrapositive of the forwards direction proves

$$v((\neg\psi \rightarrow \neg\phi)) = T \Rightarrow v(\neg(\phi \wedge \neg\psi)) = T,$$

while the contrapositive of the backwards direction proves

$$v(\neg(\phi \wedge \neg\psi)) = T \Rightarrow v((\neg\psi \rightarrow \neg\phi)) = T.$$

Hence $v(\neg(\phi \wedge \neg\psi)) = v((\neg\psi \rightarrow \neg\phi))$ for every truth assignment v , and so $\neg(\phi \wedge \neg\psi) \equiv (\neg\psi \rightarrow \neg\phi)$.

Finally, since logical equivalence is transitive by Exercise 2.35(c), Claims 1–3 chain together to give

$$(\phi \rightarrow \psi) \equiv (\neg\phi \vee \psi) \equiv \neg(\phi \wedge \neg\psi) \equiv (\neg\psi \rightarrow \neg\phi). \quad \square$$

Proof of (b). Let v be any truth assignment.

(1) Let $v((\phi \iff \psi)) = T$. Then the truth table for $\iff$ tells us that $v(\phi) = T = v(\psi)$ or $v(\phi) = F = v(\psi)$. Suppose the former is true. Then $v((\phi \rightarrow \psi)) = T$ and $v((\psi \rightarrow \phi)) = T$, and so $v(((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))) = T$. If instead $v(\phi) = F = v(\psi)$, then the antecedents of both implications are false, hence both implications take value T , and again $v(((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))) = T$.

(2) Let $v((\phi \iff \psi)) = F$. Then the truth table for $\iff$ tells us that at most one of ϕ, ψ is true and at most one is false, so the two remaining possibilities are $v(\phi) = T, v(\psi) = F$ and $v(\phi) = F, v(\psi) = T$. Swapping ϕ and ψ interchanges these two possibilities and, by the symmetry of the $\iff$ and $\wedge$ tables, changes neither side's value; so WLOG let $v(\phi) = T$ and $v(\psi) = F$. Then $v((\phi \rightarrow \psi)) = F$ and so $v(((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))) = F$.

Since v takes only the values T and F , cases (1) and (2) are exhaustive. Hence $v((\phi \iff \psi)) = v(((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)))$ for every truth assignment v , so that $(\phi \iff \psi) \equiv ((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))$. $\square$